

Functional Requirements Document (FRD) for system_name: DLT SCD2 Pipeline

1. Document Control

- Version: 1.0
- Owner:
- Last Updated:
- Source: Business Requirements Document (BRD) “Data Pipeline Steps using Delta Live Tables”

2. Scope

- In scope: Ingestion from volumes, schema enforcement, cleansing, TotalAmount derivation, join and SCD Type 2 for ordersummary, customer daily aggregate spend, DLT implementation under catalog gen_ai_poc_databrickscoe and schema sdhc_wizard.
- Out of scope: Any information not present in source JSON.

3. Actors and Systems

- Primary system: system_name: DLT SCD2 Pipeline
- Data sources: CSV in /Volumes/gen_ai_poc_databrickscoe/sdhc_wizard/customerdata and /Volumes/gen_ai_poc_databrickscoe/sdhc_wizard/orderdata
- Data targets: Delta tables in gen_ai_poc_databrickscoe.sdhc_wizard
- External actors: Data Engineer (operator) , Scheduler

4. Assumptions and Constraints

- Assumptions: DLT pipeline available; volume paths reachable; CSVs well-formed with headers .
- Constraints: Use Delta Live Tables; catalog/schema fixed; no extra logic beyond BRD; missing details marked .

5. Data Model (Logical)

- Source customer: CustId, Name, EmailId, Region; data types .
- Source order: OrderId, ItemName, PricePerUnit, Qty, Date, CustId; data types .
- Staged order_enriched: order + TotalAmount.
- Target ordersummary (SCD2): CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date, StartDate, EndDate, Status; data types .
- Target customeraggregatespend: Name, TotalAmount, Date; data types .

6. Non-Functional Requirements

- Performance: Process typical daily volume ; latency .
- Reliability: DLT managed pipeline with retries .
- Security: Access to volumes and targets via workspace IAM .

- Observability: DLT event logs and expectations .

7. Functional Requirements

- FR-ING-001 Title: Ingest customer CSV to Delta
- Description: Read customer CSV from /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata into Delta table customer using DLT.
- Preconditions: Volume path accessible; DLT cluster permissions granted.
- Main Flow:

1) Start DLT pipeline.

2) Read CSV from specified customer path.

3) Apply header and delimiter settings .

4) Write as managed Delta table customer.

- Alternate Flows:
- A1) Missing path: raise error and fail step; alert .
- Acceptance Criteria:
- AC1: Table customer created/updated in Delta.
- AC2: Row count > 0 when files present.

- FR-ING-002 Title: Ingest order CSV to Delta
- Description: Read order CSV from /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata into Delta table order using DLT.
- Preconditions: Volume path accessible; DLT cluster permissions granted.
- Main Flow:

1) Start DLT pipeline.

2) Read CSV from specified order path.

3) Apply header and delimiter settings .

4) Write as managed Delta table order.

- Alternate Flows:
- A1) Missing path: raise error and fail step; alert .
- Acceptance Criteria:
- AC1: Table order created/updated in Delta.
- AC2: Row count > 0 when files present.

- FR-SCH-003 Title: Enforce customer schema
- Description: Enforce schema CustId, Name, EmailId, Region on customer.
- Preconditions: FR-ING-001 completed.
- Main Flow:

1) Define schema with data types .

2) Cast columns to schema.

3) Handle malformed records per policy .

- Alternate Flows:
- A1) Extra/missing columns: log and apply evolution policy .
- Acceptance Criteria:
- AC1: Customer table conforms to schema fields.
- AC2: Malformed record handling aligns with policy .

- FR-SCH-004 Title: Enforce order schema
- Description: Enforce schema OrderId, ItemName, PricePerUnit, Qty, Date, CustId on order.
- Preconditions: FR-ING-002 completed.
- Main Flow:

1) Define schema with data types .

2) Cast columns to schema.

3) Handle malformed records per policy .

- Alternate Flows:
- A1) Extra/missing columns: log and apply evolution policy .
- Acceptance Criteria:
- AC1: Order table conforms to schema fields.
- AC2: Malformed record handling aligns with policy .

- FR-DQ-005 Title: Cleanse customer nulls and duplicates
- Description: Remove "Null"/Null values and duplicate records from customer.
- Preconditions: FR-SCH-003 completed.
- Main Flow:

1) Standardize literal "Null" strings to null.

2) Drop records with required column nulls (CustId, Name) .

3) Deduplicate on business key CustId keeping latest by .

- Alternate Flows:
- A1) No dedupe key available: drop exact duplicates by all columns.
- Acceptance Criteria:
- AC1: No "Null" strings remain.
- AC2: No duplicate CustId rows when business key defined; else no exact duplicates.
- FR-DQ-006 Title: Cleanse order nulls and duplicates
- Description: Remove "Null"/Null values and duplicate records from order.
- Preconditions: FR-SCH-004 completed.
- Main Flow:

1) Standardize "Null" strings to null.

2) Drop rows with nulls in OrderId, CustId, Date, PricePerUnit, Qty .

3) Deduplicate on business key OrderId keeping latest by Date .

- Alternate Flows:
- A1) If OrderId not unique: dedupe by (OrderId, ItemName, CustId, Date) .
- Acceptance Criteria:
- AC1: No "Null" strings remain.
- AC2: No duplicate OrderId rows per policy.
- FR-TRF-007 Title: Derive TotalAmount on order
- Description: Add TotalAmount = PricePerUnit × Qty.
- Preconditions: FR-DQ-006 completed.
- Main Flow:

1) Cast PricePerUnit and Qty to numeric .

2) Compute TotalAmount = PricePerUnit * Qty.

3) Persist as order_enriched view/table in DLT.

- Alternate Flows:
- A1) Non-numeric values: coerce or drop per policy .
- Acceptance Criteria:
- AC1: TotalAmount present and numeric for all retained rows.
- AC2: Sample validation aligns with manual calculation.
- FR-TBL-008 Title: Create ordersummary table

- Description: Ensure ordersummary exists in gen_ai_poc_databrickscoe.sdlic_wizard.
- Preconditions: FR-TRF-007 completed; FR-DQ-005 completed.
- Main Flow:

1) Define target with fields listed in BRD plus SCD2 fields StartDate, EndDate, Status.

2) Create managed Delta table if not exists with DLT.

- Alternate Flows:
- A1) Missing privileges: fail with actionable error.
- Acceptance Criteria:
- AC1: ordersummary exists at catalog.schema.table.
- AC2: Columns include business and SCD fields.
- FR-SCD-009 Title: Join and load to SCD2 ordersummary
- Description: Join customer and order_enriched on CustId; load into ordersummary with SCD Type 2.
- Preconditions: FR-TBL-008 completed.
- Main Flow:

1) Perform inner join customer to order_enriched on CustId.

2) Define SCD business key as (CustId, OrderId) .

3) Detect changes on mutable attributes (Name, EmailId, Region, ItemName, PricePerUnit, Qty, Date, TotalAmount) .

4) For changes: expire prior record (Status=Inactive, EndDate=current load ts).

5) Insert new active record (Status=Active, StartDate=current load ts, EndDate=null).

- Alternate Flows:
- A1) Late-arriving data: apply as-of Date handling .
- Acceptance Criteria:
- AC1: History maintained; only one active record per business key.
- AC2: StartDate/EndDate populated per change events.
- FR-AGG-010 Title: Create customeraggregatespend table
- Description: Ensure customeraggregatespend exists in gen_ai_poc_databrickscoe.sdlic_wizard with Name, TotalAmount, Date.
- Preconditions: FR-SCD-009 completed.
- Main Flow:

1) Define target schema.

2) Create managed Delta table if not exists with DLT.

- Alternate Flows:
- A1) Missing privileges: fail with actionable error.
- Acceptance Criteria:
- AC1: Table exists at catalog.schema.
- AC2: Columns Name, TotalAmount, Date present.
- FR-AGG-011 Title: Aggregate daily spend
- Description: Aggregate TotalAmount from ordersummary grouped by Name and Date into customeraggregatespend.
- Preconditions: FR-AGG-010 completed.
- Main Flow:

1) Select Name, Date, TotalAmount from ordersummary.

2) Aggregate using SUM(TotalAmount) .

3) Write results to customeraggregatespend with overwrite or merge policy .

- Alternate Flows:
- A1) If aggregation function differs: apply stakeholder-approved function .
- Acceptance Criteria:
- AC1: One row per (Name, Date).
- AC2: TotalAmount equals sum of contributing rows for the day.
- FR-OPS-012 Title: Implement pipeline in DLT
- Description: Implement all steps using Delta Live Tables with declarative expectations for DQ.
- Preconditions: Workspace and permissions ready.
- Main Flow:

1) Author DLT notebooks/pipeline specifying sources, transformations, targets.

2) Configure cluster policy .

3) Configure expectations for null and duplicate removal.

4) Schedule or trigger run .

- Alternate Flows:
- A1) Manual trigger via UI on demand.
- Acceptance Criteria:

- AC1: DLT lineage shows steps 1–7 as per BRD.
- AC2: Pipeline run succeeds with targets materialized.

8. Process Flows

- End-to-end order:
- Ingest customer and order CSV
- Cleanse nulls/duplicates
- Derive TotalAmount
- Create ordersummary
- Join + SCD2 load to ordersummary
- Create customeraggregatespend
- Aggregate daily spend

9. Error Handling

- Ingestion failure: fail-fast; log path; alert .
- Schema mismatch: quarantine records .
- SCD conflict: ensure single active record; resolve by latest Date or load ts .
- Aggregation failure: rollback write; alert .

10. Configuration and Parameters

- CSV options: header=true, delimiter=, encoding=UTF-8 .
- Timezone for dates: .
- Watermark/late data threshold: .
- Merge mode for aggregates: overwrite by Date partition .

11. Security and Access

- Read access to volumes; write access to catalog/schema; secret scope for credentials .
- Data masking for EmailId in non-prod .

12. Testing and Acceptance

- Unit tests: TotalAmount math; null/duplicate removal; SCD2 state transitions.
- Integration tests: End-to-end run with sample files.
- UAT: Business validates ordersummary history and aggregate outputs.
- Acceptance: Meet Appendix P AC-1 to AC-6.

13. Open Items

- Aggregation function confirmation (SUM assumed).

- Data types and nullable constraints.
- Dedupe keys and change-detection fields.
- SCD columns exact names and data types.
- CSV formats and schema evolution policy.
- SCD business key definition.
- Final table name confirmation for customeraggregatespend.

14. Traceability

- FR-ING-001 to FR-10; mappings per BRD Appendix R.

Mermaid diagrams

System flow diagram

[Diagram failed: Mermaid render failed (HTTP 404): <!doctype html> <html lang="en"> <head> <meta charset="utf-8"> <meta name="viewport" content="width=device-width, initial-scale=1.0"> <title>404 Not Found</title> <style> body { text-align: center; } h1 { font-size: 3em; } p { font-size: 1.25em; } </style> </head> <body> <h1>404 Not Found</h1> <hr/> <p> Please go back or return to the home page. </p> </body> </html>]

Process flow / flowchart

[Diagram failed: Mermaid render failed (HTTP 404): <!doctype html> <html lang="en"> <head> <meta charset="utf-8"> <meta name="viewport" content="width=device-width, initial-scale=1.0"> <title>404 Not Found</title> <style> body { text-align: center; } h1 { font-size: 3em; } p { font-size: 1.25em; } </style> </head> <body> <h1>404 Not Found</h1> <hr/> <p> Please go back or return to the home page. </p> </body> </html>]

Data flow diagram (DFD)

[Diagram failed: Mermaid render failed (HTTP 404): <!doctype html> <html lang="en"> <head> <meta charset="utf-8"> <meta name="viewport" content="width=device-width, initial-scale=1.0"> <title>404 Not Found</title> <style> body { text-align: center; } h1 { font-size: 3em; } p { font-size: 1.25em; } </style> </head> <body> <h1>404 Not Found</h1> <hr/> <p> Please go back or return to the home page. </p> </body> </html>]

Sequence diagram

[Diagram failed: Mermaid render failed (HTTP 404): <!doctype html> <html lang="en"> <head> <meta charset="utf-8"> <meta name="viewport" content="width=device-width, initial-scale=1.0"> <title>404 Not Found</title> <style> body { text-align: center; } h1 { font-size: 3em; } p { font-size: 1.25em; } </style> </head> <body> <h1>404 Not Found</h1> <hr/> <p> Please go back or return to the home page. </p> </body> </html>]

High-level architecture diagram

[Diagram failed: Mermaid render failed (HTTP 404): <!doctype html> <html lang="en"> <head> <meta charset="utf-8"> <meta name="viewport" content="width=device-width, initial-scale=1.0"> <title>404 Not Found</title> <style> body { text-align: center; } h1 { font-size: 3em; } p { font-size: 1.25em; } </style> </head> <body> <h1>404 Not Found</h1> <hr/> <p> Please go back or return to the home page. </p> </body> </html>]

Appendix A: Acceptance Criteria Mapping

- FR-OPS-012 satisfies AC-1.
- FR-TBL-008 satisfies AC-2.
- FR-AGG-010 satisfies AC-3.
- FR-SCD-009 satisfies AC-4.
- FR-DQ-005 and FR-DQ-006 satisfy AC-5.
- FR-TRF-007 satisfies AC-6.

Appendix B: ID Schema

- Pattern: FR--NNN where MOD $\in$ {ING,SCH,DQ,TRF,TBL,SCD,AGG,OPS}.
- Example: FR-ING-001.

Appendix C: Glossary

- DLT: Delta Live Tables.
- SCD2: Slowly Changing Dimension Type 2.
- Business key: Unique identifier for SCD history tracking .